# Beyond Pure Transformers: A Study of Hybrid Mamba-Transformer Architectures in NVIDIA's Nemotron 3 Super


## Abstract

Since 2017, the Transformer architecture has been the standard foundation for large language models. However, its self-attention mechanism becomes computationally expensive as input length increases, limiting its efficiency on long-context tasks. This seminar presents Nemotron 3 Super, a language model released by NVIDIA in April 2026, which introduces a hybrid architecture combining Mamba-based state-space layers with a smaller number of attention layers, along with a Mixture-of-Experts (MoE) design. The model has 120 billion total parameters, of which only 12 billion are active per token, allowing it to process context windows of up to 1 million tokens while achieving up to 7.5 times higher inference throughput than comparable models such as GPT-OSS-120B. The seminar will explain the working principles of this hybrid architecture, describe how it reduces computational cost without reducing accuracy, and discuss why this approach is considered a significant step beyond traditional Transformer-only designs in the development of efficient, large-scale AI systems.



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